

CMT Problem Set 3 Homework #3

Problem 3.1 (Crystallographic restriction theorem). Proof the crystallographic restriction theorem: Consider a crystal (i.e. with translational symmetry) in three dimensions that is invariant under rotations by an angle $2\pi/n$ around an axis. Then the only allowed possibilities are $n = 1, 2, 3, 4, 6$ with the case $n = 1$ being the trivial case with no rotational symmetry.

Solution. Proof. Denote the 3D lattice can be generated by the basis vectors $\vec{e}_1$, $\vec{e}_2$, and $\vec{e}_3$. So, any lattice point vector set should be

$$L = \{a_1\vec{e}_1 + a_2\vec{e}_2 + a_3\vec{e}_3\}$$

Obviously, $a_i \in \mathbb{Z}$, i.e., any lattice point can be labeled by integer coordinates (a_1, a_2, a_3) . Consider the 3D rotation matrix

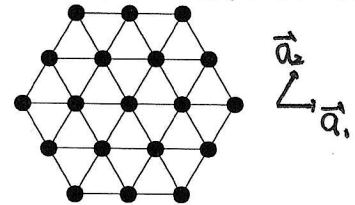
$$R = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

When acting it on the vector L , it should leave the lattice invariant, i.e., the trace must be integers

$$\text{Tr}(R) = 2\cos\theta + 1 \in \mathbb{Z}$$

So, we have $2\cos\theta = 2\cos(2\pi/n) \in \mathbb{Z}$, i.e., $\cos(2\pi/n) = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$, then $n \in \{1, 2, 3, 4, 6\}$ $\square$

Problem 3.2 (2D triangular lattice). Let us consider a two-dimensional triangular lattice as shown in following Figure.



- Identify the primitive unit cell of the two-dimensional triangular lattice. Find the basis vectors.
- Construct the basis vectors of the reciprocal unit cell.

Solution.

(a) Denote the lattice constant a . Then the simplest choice of primitive lattice vector is

$$\vec{a}_1 = a(1, 0), \quad \vec{a}_2 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

(b). Denote the two reciprocal-lattice primitives $\vec{b}_1 = (b_{1x}, b_{1y})$, $\vec{b}_2 = (b_{2x}, b_{2y})$ using the relation

$$\vec{a}_i \cdot \vec{b}_j = 2\pi \delta_{ij}$$

To expand it

$$\vec{a}_1 \cdot \vec{b}_1 = 2\pi, \quad \vec{a}_1 \cdot \vec{b}_2 = 0, \quad \vec{a}_2 \cdot \vec{b}_1 = 0, \quad \vec{a}_2 \cdot \vec{b}_2 = 2\pi$$

We obtain the two reciprocal-lattice primitive vectors

$$\vec{b}_1 = \frac{2\pi}{a}\left(1, -\frac{1}{\sqrt{3}}\right), \quad \vec{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{\sqrt{3}}\right)$$

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Problem 3.3 (Properties of the reciprocal lattice).

- (a) Show that all reciprocal lattice vectors of the form $\vec{G} = h\vec{A} + k\vec{B} + l\vec{C}$ are perpendicular to the planes of the same indices (h, k, l) in real space.
- (b) Show that the distance $d_{h,k,l}$ between two consecutive planes (h, k, l) is inversely proportional to $|\vec{G}_{h,k,l}|$.
- (c) Find $d_{h,k,l}$ for a simple cubic lattice and an orthorhombic lattice ($a \neq b \neq c, \alpha = \beta = \gamma = \pi/2$).

Solution.

(a) **Proof.** Substitute the expression of the reciprocal vectors to $\vec{G}$

$$\vec{G} = \frac{2\pi}{V} [h(\vec{a}_2 \times \vec{a}_3) + k(\vec{a}_3 \times \vec{a}_1) + l(\vec{a}_1 \times \vec{a}_2)]$$

Since the Miller indices (h, k, l) correspond to a plane that intercepts the crystal axes at $\vec{a}_1/h, \vec{a}_2/k, \vec{a}_3/l$, then, any vector lying in the (h, k, l) plane can be written as a linear combination of two independent vectors in that plane, i.e.,

$$\vec{r}_1 = \frac{\vec{a}_2}{k} - \frac{\vec{a}_1}{h}, \quad \vec{r}_2 = \frac{\vec{a}_3}{l} - \frac{\vec{a}_2}{k}$$

Then, the normal to this plane can be written as a cross product

$$\vec{n}_{hkl} = \vec{r}_1 \times \vec{r}_2 = \frac{1}{kl}(\vec{a}_2 \times \vec{a}_3) - \frac{1}{hl}(\vec{a}_1 \times \vec{a}_3) + \frac{1}{hk}(\vec{a}_1 \times \vec{a}_2) = \frac{1}{kl}(\vec{a}_2 \times \vec{a}_3) + \frac{1}{hl}(\vec{a}_3 \times \vec{a}_1) + \frac{1}{hk}(\vec{a}_1 \times \vec{a}_2)$$

which is parallel to $\vec{G}$, implies that $\vec{G}$ is perpendicular to the (h, k, l) plane.

(b) **Proof.** For the vector pointing to the $n+1$ and n -th (h, k, l) plane, it satisfies

$$\vec{G} \cdot \vec{r}_{n+1} = 2\pi(n+1), \quad \vec{G} \cdot \vec{r}_n = 2\pi n.$$

where $\vec{r}_{n+1} - \vec{r}_n = d \cdot \hat{n}$. Combine the two relations

$$\vec{G} \cdot (\vec{r}_{n+1} - \vec{r}_n) = \vec{G} \cdot \hat{n} \cdot d = 2\pi d.$$

So, we have

$$d = \frac{2\pi}{\vec{G} \cdot \hat{n}} = \frac{2\pi}{|\vec{G}_{hkl}|}$$

i.e., the distance between two consecutive planes is inversely proportional to $|\vec{G}_{hkl}|$

(c) For a simple cubic lattice $\vec{A} = (a, 0, 0)$, $\vec{B} = (0, b, 0)$, $\vec{C} = (0, 0, c)$, the reciprocal vectors can be expressed as

$$\vec{G} = \frac{2\pi h}{a} \hat{x} + \frac{2\pi k}{b} \hat{y} + \frac{2\pi l}{c} \hat{z}$$

and its magnitude $|\vec{G}| = 2\pi \sqrt{\left(\frac{h}{a}\right)^2 + \left(\frac{k}{b}\right)^2 + \left(\frac{l}{c}\right)^2}$, so we obtain the distance

$$d = \frac{2\pi}{|\vec{G}|} = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right)^{-\frac{1}{2}}.$$

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Problem 3.4 (Atomic planes and Miller indices: application to lithium). The Bravais lattice of lithium is simple cubic with lattice parameter $a = 3.48 \text{ \AA}$.

- Suppose that the atoms (assumed to be spheres) are placed along the $[111]$, represent the atomic distribution along the following faces (100) , (110) , (111) , and (201) .
- For each such 2D structure, state the direction and basis vector, $\vec{a}$ and $\vec{b}$ of the elementary lattice as well as the value of the angle, γ .
- Find the atomic concentration and the mass density of lithium ($A \approx 7$).

Solution.

(a) (b) The distribution, basis vector, and the angle of the faces are listed

2D Structure	Distribution	$\vec{a}$	$\vec{b}$	γ
(100)	Square	$(0, a, 0)$	$(0, 0, a)$	90°
(110)	Rectangular	$(a, -a, 0)$	$(0, 0, a)$	90°
(111)	Hexagonal	$(a, 0, -a)$	$(0, a, -a)$	60°
(201)	Rectangular	$(0, a, 0)$	$(a, 0, -2a)$	90°

(c) Since the atoms are placed along $[111]$, then every cubic lattice's corners has an atom, i.e., equivalent that there is an atom in every lattice. So, the atomic concentration is

$$n = \frac{1}{V} = \frac{1}{a^3} = 2.3728 \times 10^{28} \text{ atoms/m}^3$$

The mass density is

$$\rho = n m_{\text{atom}} = \frac{nA}{N_A} = 275.816 \text{ kg/m}^3$$